

Cybersecurity Training Diary

Day 25: Cyber Threat Simulation & Red Team Operations

1. Topics Covered

✓ Red Team vs Penetration Testing

```
graph LR
  A[PenTest] -->|Focused| B[Find Vulnerabilities]
  C[Red Team] -->|Stealthy| D[Simulate Advanced Adversaries]
```

✓ Attack Simulation Framework

Phase	Objective	Tools
Initial Access	Gain foothold	Cobalt Strike, Mythic
Persistence	Maintain access	Empire, Sliver
Lateral Movement	Expand control	Mimikatz, BloodHound
Exfiltration	Data theft	Rclone, DNSCat2

✓ Purple Team Exercises

- **Collaborative** red/blue team drills
- **Automated** feedback loops (Caldera, Atomic Red Team)

2. Key Takeaways

- ✓ **Red teaming is about emulation** - think like APT groups (MITRE ATT&CK mapped)
- ✓ **Purple teaming maximizes learning** - immediate defensive feedback
- ✓ **OPSEC failures** get red teams caught (unusual traffic, tool signatures)

3. Practical Exercises

- ♦ **Exercise 1: Simulate Phishing Campaign**

```
# With GoPhish
./gophish-admin create-campaign \
  --template "urgent_reset.html" \
  --targets employees.csv
```

- ♦ **Exercise 2: Bypass AV with Obfuscation**

```
# PowerShell Empire stager
$enc = [System.Convert]::ToBase64String([System.Text.Encoding]::Unicode.GetBytes($code))
iex ([System.Text.Encoding]::Unicode.GetString([System.Convert]::FromBase64String($enc)))
```

4. Questions for Reflection

- ? How often should organizations run red team exercises?
- ? What separates elite red teamers from script kiddies?